

Nicolas Chatzikiriakos

 Institute for Systems Theory and Automatic Control, University of Stuttgart

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About me

I am a PhD student at the Institute for Systems Theory and Automatic Control at the University of Stuttgart, where I am advised by Andrea Iannelli. My main research is leveraging statistical learning theory tools for data-driven control. With this, I am particularly interested in quantifying the uncertainty caused by noise in the data and analyzing the statistical hardness of learning dynamical systems.

Education

PhD	Institute for Systems Theory and Automatic Control, University of Stuttgart	Since Mai 2023
	• Advisor: Andrea Iannelli • Research interest: Uncertainty quantification for data-driven control	
M.Sc.	University of Stuttgart , Engineering Cybernetics	Mar. 2021 – Apr. 2023
	• Final Grade: 1.3 • Thesis: <i>Safe approximation of model predictive controllers using neural networks</i> (in cooperation with Robert Bosch GmbH) • Coursework: Robust Control, Optimal control, Model Predictive Control, Data-driven Control	
B.Sc.	University of Stuttgart , Engineering Cybernetics	Oct. 2017 – Mar. 2021
	• Final Grade: 1.7 • Thesis: Microscopic modelling and simulation of German highway traffic with regards to string stable cruise control	

Research Stays

Allen School of Computer Science & Engineering	University of Washington, Seattle, WA, USA
• Host: Prof. Kevin Jamieson • Topics: Stochastic Bandits, Experimental Design, Learning in stochastic environments	Mar 2025 – Jul 2025

Experience

Robert Bosch GmbH , Internship	Research Campus, Renningen, Germany
• Development of a direct-switching model predictive control method for the efficient operation of electrical machines • Development of a modular design of the control structure and its implementation in Matlab/Simulink • Transfer of the control structure from the simulation environment to Rapid Control prototyping environment (dSPACE) • Validation of the robustness of the controller on the test bench	Oct 2021 – Mar 2022

Teaching

Led the **Restructuring of the Student Laboratory** Concepts of Automatic Control
Teaching Assistant Introduction to Automatic Control
Organization of IST Honours Course
Student Laboratory Concepts of Automatic Control
Student Laboratory Introduction to Automatic Control

Summer Term 26
Since Winter Term 25
Since Winter Term 24
Winter Terms 23 & 24
Summer Terms 23 & 24

Accepted Publications

End-to-end guarantees for indirect data-driven control of bilinear systems with finite stochastic data

Automatica 2026

Chatzikiriakos, N., Strässer, R., Allgöwer, F., Iannelli, A.

Automatica 187 (2026): 112908



Hidden Convexity in Active Learning: A Convexified Online Input Design for ARX Systems

CDC 2025

Chatzikiriakos, N., Song, B., Rank, P., Iannelli, A.

Proceeding of 64th IEEE Conference on Decision and Control



Sample Complexity Bounds for Linear System Identification From a Finite Set

L-CSS (ACC 2025)

Chatzikiriakos, N., Iannelli, A.

IEEE Control Systems Letters (Volume: 8)



Learning soft constrained MPC value functions: Efficient MPC design and implementation providing stability and safety guarantees

L4DC 2024

Chatzikiriakos, N.*, Wabersich, K.P.*, Berkel, F., Pauli, P., Iannelli, A.

Proceedings of the 6th Annual Learning for Dynamics & Control Conference 2024



Skills

Languages: German (mother tongue), English (C1), French (B1), Spanish (A2)

Programming Languages: Matlab, Python

Reviewer: ACC, L4DC, CDC, EWRL, JMLR